1. What status codes DevOps engineers deal with

DevOps engineers mainly work with **HTTP status codes (1xx–5xx)** while troubleshooting:

🔹 **1xx – Informational (rarely used)**

- 100 Continue
  👉 Request received, continue sending body
  ✔️ Mostly internal, not used in debugging

- 🔹 **2xx – Success**

Code	Meaning	DevOps relevance
200	OK	Everything working
201	Created	Resource created (POST API)
204	No Content	Success but no response

- 👉 Usually no issue — confirms system is healthy
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3xx – Redirection

Code	Meaning	DevOps relevance
301	Permanent redirect	DNS / domain / HTTPS redirect
302	Temporary redirect	App-level redirect
304	Not Modified	Cache working

👉 Important for:

- HTTP → HTTPS redirect
- Domain mapping issues

4xx – Client Errors (VERY IMPORTANT)

These mean **request issue from client/frontend/API consumer**

Code	Meaning	Common DevOps cause
400	Bad Request	Wrong payload
401	Unauthorized	Missing token
403	Forbidden	Permission issue
404	Not Found	Wrong route / Nginx config
405	Method Not Allowed	Wrong HTTP method
408	Timeout	Slow client
429	Too Many Requests	Rate limiting

5xx – Server Errors (CRITICAL for DevOps)

These mean **backend/server issue**

Code	Meaning	DevOps impact
500	Internal Server Error	App crash
502	Bad Gateway	Nginx can't reach backend
503	Service Unavailable	Server down / overload
504	Gateway Timeout	Backend too slow

2. How DevOps Engineers Think (Golden Rule)

👉 **4xx → Client problem**
👉 **5xx → Server problem (YOUR responsibility)**

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3. Real-world Troubleshooting (Very Important)

Let's go one by one (practical scenarios 👇)

404 Not Found 📌 **Meaning:** Requested URL not found 🔍 **Causes:** • Wrong API route • Nginx config issue • Static files missing 🛠 **Troubleshooting:** # Check nginx config cat /etc/nginx/sites-enabled/default # Check file exists ls /var/www/html # Test backend curl http://localhost:3000/api	**502 Bad Gateway** 👉 This is **MOST COMMON DevOps issue** 📌 **Meaning:** Nginx / Load Balancer can't connect to backend 🔍 **Causes:** • Backend not running • Wrong port • Security group blocked 🛠 **Troubleshooting:** pm2 status netstat -tulnp | grep 3000 # test backend directly curl http://localhost:3000

503 Service Unavailable 📌 **Meaning:** Server overloaded or down 🔍 **Causes:** • App crashed • Too many requests • Auto scaling not working 🛠 **Troubleshooting:** top free -m df -h pm2 logs	**504 Gateway Timeout** 📌 **Meaning:** Backend taking too long 🔍 **Causes:** • Slow DB query • API delay • Network latency 🛠 **Troubleshooting:** # increase timeout in nginx proxy_read_timeout 60; # check logs tail -f /var/log/nginx/error.log
500 Internal Server Error 📌 **Meaning:** Application error 🔍 **Causes:** • Code bug • Unhandled exception • DB connection issue 🛠 **Troubleshooting:** pm2 logs docker logs <container> journalctl -u app.service	**403 Forbidden** 📌 **Meaning:** Access denied 🔍 **Causes:** • File permission issue • Security group • IAM / auth issue 🛠 **Troubleshooting:** ls -l /var/www/html chmod -R 755 /var/www/html chown -R www-data:www-data /var/www/html

401 Unauthorized	429 Too Many Requests
📌 **Meaning:** Authentication required 🔍 **Causes:** • Missing token • Expired JWT 🛠 **Troubleshooting:** • Check headers • Validate token • Check auth service	📌 **Meaning:** Rate limit exceeded 🔍 **Causes:** • DDoS or heavy traffic • API rate limiting 🛠 **Troubleshooting:** • Check Nginx: limit_req_zone • Use caching / CDN • Scale system

4. DevOps Tools Used for Debugging

🔧 **Logs**

- Nginx:

/var/log/nginx/access.log

/var/log/nginx/error.log

- App:

pm2 logs

docker logs

🔧 **Network Debugging**

curl -I https://yourdomain.com

telnet localhost 3000

🔧 **AWS Level**

- Load Balancer health checks
- CloudWatch logs
- Auto Scaling metrics

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5. End-to-End Debugging Flow (VERY IMPORTANT)

When issue comes:

Step-by-step thinking:

1. **Check status code**
2. If 4xx → check frontend/request
3. If 5xx → check backend

Real DevOps Flow:

User → Domain → Load Balancer → Nginx → Backend → DB

👉 Check layer by layer:

- DNS issue?
- SSL issue?
- Nginx issue?
- Backend down?
- DB slow?

✅ **6. Interview Tip (Important)**

If asked:

👉 "Which status codes DevOps handles?"

Answer:

- Mainly **5xx errors**
- Also **502, 503, 504 (most critical)**
- Sometimes **404, 403 (infra/config issues)**